

Special Issues

Project Information

- Project: Project 1 Part 2
- Application: FitCheck
- Cloud Platform: AWS Academy Learner Lab
- AWS Service: Amazon EC2
- Operating System: Ubuntu
- Repository: [FitCheck Project 1 Part 2](#)

Current Deployed Website

The FitCheck application was deployed on an Amazon EC2 Ubuntu instance using Docker.

Deployed Website: [Open FitCheck Website](#)

Introduction

This page explains the behavior of Amazon EC2 instances when they are stopped, rebooted, restored, or assigned a public IP address. It also discusses AMIs, EBS snapshots, Elastic IP addresses, AWS costs, and the availability of these features in the AWS Academy Learner Lab.

The information is based on AWS EC2 documentation and the standard behavior of EC2 resources.

Special Issue 1: What Happens When an EC2 Instance Stops Running?

Difference Between Stopping and Terminating

Stopping an EC2 instance is different from terminating an EC2 instance.

When an instance is stopped, the instance can normally be started again. When an instance is terminated, the instance is permanently deleted and may not be recoverable.

Action	Result
Reboot	Restarts the operating system
Stop	Shuts down the instance but keeps the EBS volume
Start	Starts a previously stopped instance
Terminate	Permanently deletes the instance

What Happens When an Instance Is Stopped?

When an EC2 instance is stopped:

- The operating system shuts down.
- Data stored in RAM is lost.
- Running processes are stopped.
- Temporary instance-store data may be lost.
- Data stored on Amazon EBS volumes normally remains.
- Installed applications and files on the EBS root volume normally remain.
- The automatically assigned public IPv4 address may change when the instance is started again.
- EC2 compute charges normally stop while the instance is stopped.
- EBS storage charges continue while the EBS volume exists.

For the FitCheck project, the source code and Docker files stored on the EBS volume would normally remain after a stop and start operation. However, unsaved information in memory would be lost.

How Can the Instance Be Protected?

The instance can be protected by creating backups before stopping or terminating it.

The available backup options include:

1. Creating an Amazon Machine Image.
2. Creating an EBS snapshot.
3. Saving application files in GitHub.
4. Copying important files to Amazon S3.
5. Using an Elastic IP address to preserve a public address.

Creating an Amazon Machine Image

An Amazon Machine Image, or AMI, is a reusable copy of an EC2 instance. An AMI can contain:

- The Ubuntu operating system.
- Installed software.
- Application files.
- Docker configuration.
- EBS volume data.
- Server configuration.

An AMI can be used to create a new EC2 instance with the same configuration as the original instance.

The general process is:

1. Open the EC2 Console.
2. Select the EC2 instance.
3. Choose Actions.
4. Choose Image and templates.
5. Select Create image.
6. Enter an AMI name and description.
7. Create the image.

8. Use the AMI later to launch a new instance.

Restoring or Duplicating an Instance

After an AMI is created, it can be used to restore or duplicate the instance.

The AMI can be selected from the EC2 Console and used to launch a new instance. The new instance can contain the same operating system, application files, installed software, and Docker configuration.

This is useful if:

- The original instance is damaged.
- The instance is accidentally terminated.
- A second copy of the application is needed.
- The application needs to be moved to another EC2 instance.

EBS Snapshots

An EBS snapshot is a point-in-time backup of an EBS volume. It can be used to restore data or create a new EBS volume.

An EBS snapshot can protect:

- Application files.
- Docker files.
- Configuration files.
- Website data.
- Operating system data stored on the EBS volume.

EBS snapshots are stored and managed by AWS. They can be used to recreate an EBS volume later.

Costs Involved

The possible costs for this solution include:

AWS Resource	Cost
Running EC2 instance	Compute charges while the instance is running
Stopped EC2 instance	No normal compute charge, but attached EBS storage can still be charged
EBS volume	Storage charges continue while the volume exists
EBS snapshot	Snapshot storage charges may apply
AMI	AMI-related EBS snapshots may create storage charges
Amazon S3	Storage and data transfer charges may apply
Public IPv4 address	Public IPv4 charges may apply

The exact costs depend on the AWS Region, the amount of storage, and the current AWS pricing.

To reduce charges, unused EC2 instances, EBS volumes, snapshots, AMIs, and Elastic IP addresses should be deleted or released.

Answer to Special Issue 1

When an EC2 instance stops, RAM data and temporary instance-store data can be lost. Data stored on EBS volumes normally remains. The instance can be started again, but its automatically assigned public IPv4 address may change.

The instance can be protected by creating an AMI, EBS snapshot, GitHub backup, or Amazon S3 backup. An AMI can be used later to restore or duplicate the original instance.

Special Issue 2: What Happens When an EC2 Instance Is Rebooted?

Reboot Behavior

A reboot restarts the operating system on the same EC2 instance.

When an instance is rebooted:

- The operating system restarts.
- The EC2 instance remains available.
- EBS data remains.
- The private IPv4 address remains.
- The public IPv4 address normally remains the same.
- Running processes stop and start again.
- Docker containers may need to be restarted.
- EC2 billing continues during the reboot.

A reboot is similar to restarting a normal computer.

Reboot Versus Stop and Start

The public IP address usually changes after a stop and start, not after a normal reboot.

Operation	Public IP Address
Reboot	Normally remains the same
Stop and start	Automatically assigned public IPv4 may change
Terminate and create a new instance	A new public IPv4 address is normally assigned
Elastic IP attached	A stable public IPv4 address can be used

Therefore, if the instance is only rebooted, the FitCheck website URL normally continues to use the same address. If the instance is stopped and started, the public IP address may change.

Why Could the Website URL Change?

The current FitCheck website is available at:

Current Website URL: [Open FitCheck Website](#)

If an EC2 instance is stopped and started again, AWS may assign a different public IPv4 address. In that situation, the previous IP address may no longer open the website.

For example:

```
Previous address: http://OLD_PUBLIC_IP  
Possible new address: http://NEW_PUBLIC_IP
```

The application files may still be available, but the old IP address may no longer point to the EC2 instance.

Options for Publishing a Stable URL

Option 1: Elastic IP Address

An Elastic IP address is a static public IPv4 address that can be associated with an EC2 instance.

Using an Elastic IP allows the website to keep the same public address after the instance is stopped and started.

Example:

```
http://YOUR_ELASTIC_IP
```

The general process is:

1. Open the EC2 Console.
2. Select Elastic IPs.
3. Choose Allocate Elastic IP address.
4. Select the new Elastic IP.
5. Choose Actions.
6. Select Associate Elastic IP address.
7. Select the EC2 instance.
8. Associate the address with the instance.

Option 2: Domain Name

A domain name can be connected to an Elastic IP address using DNS.

Example:

```
www.example.com
```

The domain name points to the Elastic IP address, allowing users to access the website using a consistent URL.

A domain name was not required for this project. The EC2 public IPv4 address was used to demonstrate the deployed FitCheck website.

Option 3: Update the Website URL

If an Elastic IP cannot be used, the new public IPv4 address can be copied from the EC2 Console after the instance starts.

The GitHub Wiki, PDF, and project documentation can then be updated with the new address.

Fees for Elastic IP and Public IPv4 Addresses

AWS charges for public IPv4 addresses. These charges can apply to:

- Automatically assigned public IPv4 addresses.
- Elastic IP addresses associated with resources.
- Elastic IP addresses that are not associated with a resource.
- Elastic IP addresses associated with stopped resources.

The exact price depends on the AWS Region and the current AWS pricing.

Unused Elastic IP addresses should be released to avoid unnecessary charges.

Can This Be Done in AWS Academy Learner Lab?

The availability of AMI creation, EBS snapshots, and Elastic IP addresses depends on the permissions and limits configured for the AWS Academy Learner Lab.

These features can be used when they are enabled in the Learner Lab account. The AWS Console should be checked to confirm the available options, service limits, and any restrictions.

The Learner Lab may allow:

- Creating an EC2 instance.
- Rebooting an EC2 instance.
- Stopping and starting an EC2 instance.
- Creating an AMI.
- Creating EBS snapshots.
- Allocating an Elastic IP address.

However, the Learner Lab may restrict:

- Some AWS services.
- Elastic IP addresses.
- AMI creation.
- Instance types.
- The number of EC2 instances.
- AWS Regions.
- Available lab credits.

The availability of these features depends on the permissions and restrictions configured in AWS Academy Learner Lab.

Answer to Special Issue 2

A normal reboot restarts the operating system and normally keeps the same public IP address. A stop-and-start operation may assign a new public IPv4 address.

To publish the website using a stable address, an Elastic IP can be associated with the EC2 instance. A domain name can also be connected to the Elastic IP. Public IPv4 and Elastic IP charges may apply.

The ability to use these features in AWS Academy Learner Lab depends on the permissions and restrictions configured for the lab.

AWS References

- [EC2 Stop and Start Behavior](#)
- [EC2 Instance State Changes](#)
- [Creating an EBS-Backed AMI](#)
- [Amazon EBS Snapshots](#)
- [EC2 Instance IP Addressing](#)
- [Elastic IP Addresses](#)
- [AWS Public IPv4 Pricing](#)

Conclusion

The special issues were answered through AWS documentation and analysis of EC2 behavior.

The main conclusions are:

- Stopping an EC2 instance normally preserves EBS data.
- RAM and temporary instance-store data can be lost.
- An AMI or EBS snapshot can be used to restore or duplicate an instance.
- A normal reboot normally keeps the same public IP.
- Stopping and starting an instance can change its public IP.
- An Elastic IP can provide a stable public address.
- AWS charges may apply for EBS storage, snapshots, public IPv4 addresses, Elastic IP addresses, and S3 storage.
- AWS Academy Learner Lab permissions determine whether AMI, snapshot, and Elastic IP operations are available.
- These special issues were researched but were not directly tested in the Learner Lab for this project.